

Om Patel

SOFTWARE ENGINEER

Python | Java | SQL | Flask | REST APIs | MySQL | AWS | Docker | Git | LLM Integration
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SUMMARY

Software engineer (B.S. Computer Science, May 2026) who designs and ships end-to-end backend and full-stack systems: REST APIs, relational database schemas, cloud deployments on AWS, and LLM-integration pipelines. Hands-on with Python, Java, and SQL across fintech transaction processing, clinical document summarization, and ML-powered matching engines. Seeking entry-level software engineering roles where I can own features from design through production deployment.

TECHNICAL SKILLS

Languages: Python | Java | SQL | JavaScript | HTML/CSS
Backend & APIs: Flask | REST API Design | Microservices | Session Management | Batch Processing | Real-Time Data Processing
Databases: MySQL | Relational Schema Design (11-table schema) | Audit Trails | ETL & Data Ingestion
Cloud & DevOps: AWS (EC2, VPC, Security Groups) | Nginx | Gunicorn | systemd | Docker | Git/GitHub | Linux | CI/CD | Agile/Scrum
AI/ML & LLM Integration: Groq (LLaMA 3.3), Gemini APIs | Prompt Engineering | Document Parsing (pdfplumber) | scikit-learn | Model Evaluation

EDUCATION

Bachelor of Science in Computer Science, Minor in Mathematics

May 2026

Roosevelt University, Chicago, IL | Major GPA: 3.63/4.0

WORK EXPERIENCE

Software Engineer Intern – Zetheta Algorithms, Remote

Jun 2026-Present

- Designed high-throughput distributed fintech transaction processing system targeting 12,000+ TPS, covering API gateway architecture, PostgreSQL + Citus sharding, Kafka event streaming, and Redis caching
- Authored OpenAPI specification and REST API gateway design for high-volume transaction streams from partner banks, with saga-pattern concurrency control and idempotency handling
- Engineered fault-tolerance strategy (circuit breakers, FMEA analysis) and load-testing plan validated across three capacity scales (1.2K / 12K / 24K TPS)
- Built capacity-planning cost model in Excel with live formulas, confirming the 12K TPS design fits within a \$45K/month infrastructure budget with 9.2% headroom

AI Intern – Coding Samurai, Remote

Apr 2026 – May 2026

- Built rule-based chatbot in Python with conditional logic and dynamic response generation
- Developed stock price forecasting model using Linear Regression with real-time market data (yfinance) and engineered features
- Implemented evaluation-metrics visualization for model performance analysis

Medical Assistant & Front Desk – Suburban Medical Wellness Center, IL

May 2025 – Sep 2025

- Designed and shipped Excel macros automating insurance-verification and appointment-confirmation workflows, saving 5+ hours per week for a 4-person front-desk team
- Managed daily check-in workflows for 50+ patient visits in eClinicalWorks EHR, maintaining accuracy across scheduling, insurance verification, and electronic health records
- Coordinated patient flow and clinical support across 3 providers and 10+ appointments per shift

Member & Developer – Roosevelt University Robotics Club, Chicago, IL

Jan 2025 – Apr 2025

- Programmed 2 prototype robots in Java and Python on microcontrollers, developing a custom obstacle-avoidance algorithm validated across campus competition runs
- Authored technical documentation and delivered onboarding presentations to a 20-person team

PROJECTS

MediSumAI – Clinical Notes Summarizer – Flask, Groq, AWS EC2, Nginx

Apr 2026 – Jul 2026

- Developed Flask web app leveraging Groq API (LLaMA 3.3 70B) to convert unstructured clinical notes into structured summaries across 4 specialties, cutting manual summarization time by ~50%
- Deployed to production on AWS EC2 (Ubuntu) with Nginx reverse proxy, Gunicorn WSGI server, and systemd service management for automatic restart and boot persistence; configured VPC security groups for least-privilege network access
- Designed and tuned LLM prompts; implemented batch processing for concurrent summarization with 5 structured output sections (Chief Complaint, HPI, Key Findings, Assessment, Plan)
- Architected ReportLab PDF export pipeline and session-history sidebar; managed API secrets via environment variables and published to GitHub

Live demo: <http://18.190.158.149> | **Code:** github.com/OmPatel0718/MediSumAI

Personal Portfolio Website – Flask, HTML/CSS/JS, Render

Jul 2026

- Designed and deployed a responsive single-page portfolio (Flask on Render) with scroll-reveal animations, live project showcases, and GitHub/contact integration; maintained via Git with continuous deployment

Transfer Credit Evaluation System – Python, MySQL, Docker, Gemini API

Feb 2026 – May 2026

- Built end-to-end PDF transcript parser with pdfplumber and a Gemini 1.5 Flash AI fallback layer, ingesting structured course records from multiple universities
- Engineered Jaccard similarity matching engine mapping 25 courses to 26 CAE-aligned Knowledge Units, achieving 85–90% accuracy across 4 institutions
- Implemented 3-tier classifier (Full/Partial/No Transfer) with credit-hour validation, confidence thresholds, and a MySQL audit trail across 11 relational tables; containerized with Docker

CERTIFICATIONS & TRAINING

- Deloitte Technology Job Simulation — Python JSON unification & software development proposal (Forage, 2026)
- Deloitte Data Analytics Job Simulation (Forage, 2026)
- Google AI Essentials (2026)
- Anthropic Academy — Claude 101 & AI Fluency (2026)